

Notes: Heath and Mobarak (JDE, 2015)

Manufacturing growth and the lives of Bangladeshi women

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1 Big picture

- **Question.** What happens to girls' schooling, labor supply, marriage, and childbearing when export-oriented manufacturing jobs suddenly become available nearby?
- **Setting.** The paper studies the rapid expansion of the Bangladeshi ready-made garment industry and compares girls in villages within commuting distance of factories to girls in nearby comparison villages, while also exploiting the timing of when garment work first became accessible in each village.
- **Core challenge.** Garment factories do not locate randomly. Villages near factories are closer to Dhaka and differ in some baseline characteristics, so the paper needs a design that separates the effect of garment-job access from pre-existing village differences and broad time trends.
- **Main contribution.** The paper shows that manufacturing growth can improve women's outcomes through both a **work channel** and an **human-capital channel**. Access to garment jobs delays marriage and childbirth, raises girls' schooling, and increases female labor force participation.
- **Why it matters.** The paper shifts attention away from purely supply-side education policies and shows that **labor demand** can be a powerful force behind girls' schooling and delayed marriage. It also matters for policy debates around anti-sweatshop boycotts: restricting garment exports may hurt the very women those policies aim to protect.
- **Main empirical design.** The authors combine:
 1. cross-village differences between garment-proximate and non-garment villages,
 2. within-garment-village variation in the **date of first local garment exposure**, and
 3. within-family comparisons between girls and their brothers in some specifications.
- **Main empirical results.**
 - Girls exposed to garment jobs are more likely to have worked outside the home; exposure during ages 10–29 raises the probability by about **12 percentage points**, and exposure during ages 10–23 by about **13 percentage points**.
 - Girls with average exposure of 6.4 years are about **0.3 percentage points** less likely to marry in a given year, which is roughly a **28% decline** relative to the mean marriage hazard.
 - The same exposure lowers the first-birth hazard by about **0.23 percentage points**, around a **29% decline** relative to the mean hazard.
 - Marriage-delay effects are concentrated at ages **12–18**, especially the youngest ages.
 - Each additional year of garment exposure raises boys' schooling by about **0.26 years** and girls' schooling by an **additional 0.22 years**; at average exposure this closes the gender education gap by about **1.5 years**.
 - Girls' school enrollment rises strongly at younger ages; for example, an 8-year-old girl in a garment-proximate village is about **13 percentage points** more likely to be enrolled after factories arrive, relative to comparable girls in non-garment villages.
 - The widely discussed female school stipend program does **not** show comparable effects on enrollment, marriage, or childbearing in this design.

- **Economic takeaway.** Manufacturing growth changed girls’ opportunity sets in Bangladesh. It raised the returns to education for young girls and raised the opportunity cost of early marriage for older girls. The result is a combination of **more schooling**, **more market work**, and **later marriage and fertility**.

2 Data and measurement (key objects)

2.1 Survey, sample, and location

- **Main data.** The authors conducted a survey of **1,395 households** in **60 villages** in four sub-districts near Dhaka: Savar and Dhamrai in Dhaka district, and Gazipur Sadar and Kaliakur in Gazipur district.
- **Timing.** The core household survey was conducted in **August–September 2009**. The authors revisited villages in **March 2014** to collect supplemental information on the dates when nearby garment employment first became available.
- **Village classification.** A garment manufacturers’ association official classified **44 villages** as within commuting distance of garment factories and **16 villages** as not within commuting distance.
- **Validation of treatment status.** In garment-proximate villages, **39%** of women ages 16–30 work in garment factories, compared with only **2%** in control villages.
- **Why the peri-urban setting matters.** These are villages outside Dhaka city but affected by urban expansion and industrial growth. This gives the paper a setting where factories are reachable by commuting rather than requiring permanent migration.

2.2 Retrospective histories and unit of observation

- **Household-level histories.** The survey records schooling, marriage, and childbirth histories for all offspring of the household head and spouse, including children no longer residing in the household.
- **Key retrospective measures.**
 - age school started,
 - interruptions in schooling,
 - completed years of education,
 - age at marriage,
 - age at first birth,
 - whether female offspring ever worked outside the home.
- **Panel-like reconstruction.** These retrospective histories allow the authors to construct person-year outcomes, such as whether a child aged 5–18 was enrolled in school in year t , or whether a girl married in year t .

2.3 Key treatment variables

- **GarmentVillage.** Indicator equal to one if the origin village is within commuting distance of garment factories.
- **YearsExposure.** A continuous measure counting how many years, during a relevant period, someone from the individual's origin village had already been working in the garment industry.
- **PostGarments.** Indicator equal to one if garment work had become available to that village by that year.
- **AnyExposure / critical-period exposure.** Binary indicators for whether a person was exposed to garment jobs during key ages, such as ages 10–29 for work decisions or ages 10–23 for marriage-relevant exposure.
- **Why timing matters.** Two girls in the same garment village can have different exposure depending on their cohort: one may have grown up mostly before factory access arrived, while another may have spent her whole childhood after access arrived.

2.4 Background facts on the garment sector and workers

- **National growth.** The garment sector was virtually nonexistent in 1980, employed around 40,000 workers in 1983, and grew at about **17% per year** to roughly **4 million workers** by the early 2010s.
- **Female employment.** Nationally, about **80%** of garment workers are female, and the authors estimate that around **15%** of Bangladeshi women ages 16–30 work in the garment industry.
- **Worker profile in the sample.** The middle 50% of sampled female garment workers are roughly between ages **19 and 28**, but about **20%** are 18 or younger.
- **Education of workers.** More than half of sampled female garment workers have **5 or more years of education**. In the survey, garment workers average about **6.1 years** of schooling versus **4.9** for other workers in garment villages and **3.3** in non-garment villages.
- **Gender composition in this sample.** The local sample has more woven factories than the national industry, so only **54%** of sampled garment workers are female. But the paper argues the jobs available to girls are still very similar to the broader garment sector.
- **Evidence on returns to skill.**
 - Conditional on age and experience, an extra year of schooling is associated with about **3.7%** higher wages among workers.
 - Conditional on age, an extra year of schooling is associated with about a **2.1%** higher probability of working in the garment industry.

2.5 Baseline differences and descriptive concerns

- **Distance to Dhaka.** Garment-proximate villages are much closer to Dhaka on average: about **1.8 km** versus **6.8 km**.

- **Older cohorts.** Adults over 50 in garment-proximate villages have somewhat higher schooling, especially men, suggesting these villages were already somewhat different before garment takeoff.
- **No obvious school-access difference.** Distance to girls' and boys' secondary schools and male/female agricultural wages look similar across treatment and control villages.
- **Implication.** Identification cannot rely on a simple treated-versus-control comparison alone; it must rely on **timing variation**, **trend controls**, and **within-family comparisons**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Female labor force participation

The paper first studies whether garment-job access actually moved girls into work outside the home. For girl i in village v :

$$\text{EverWorked}_{iv} = \delta \text{GarmentVillage}_v \times \text{GarmentExposure}_{iv}^{(t_0-t_1)} + \gamma \text{GarmentVillage}_v + \varepsilon_{iv}.$$

- EverWorked_{iv} is an indicator for whether the girl ever worked outside the home.
- $\text{GarmentExposure}_{iv}^{(t_0-t_1)}$ indicates exposure during a critical age range, such as ages 10–29 or 10–23.
- δ captures how much more likely girls in garment villages are to work when garment access arrives during those critical years.

Interpretation. This is a reduced-form difference-in-differences specification. The key object is whether factory access coincides with the ages at which girls are likely to choose between schooling, work, and marriage.

3.2 Marriage and childbearing hazard models

The paper next estimates discrete-time hazard models for marriage and first birth. For girl i in village v and year t :

$$\begin{aligned} 1(\text{Married})_{ivt} = & \delta \text{GarmentVillage}_v \times \text{YearsExposure}_{ivt} + \gamma_1 \text{GarmentVillage}_v \\ & + \gamma_2 \text{GarmentVillage}_v \times \text{Year}_t + \beta \text{Age}_{ivt} + \lambda_t + \varepsilon_{ivt}. \end{aligned}$$

An analogous specification is estimated with $1(\text{FirstBirth})_{ivt}$ as the dependent variable.

- $1(\text{Married})_{ivt}$ equals one in the year a girl marries, and zero beforehand.
- $1(\text{FirstBirth})_{ivt}$ equals one in the year of first birth, and zero beforehand.
- $\text{YearsExposure}_{ivt}$ counts cumulative exposure to garment jobs from birth up to year t .

The authors also estimate versions interacting exposure with age and age-squared to locate the ages at which the hazard effects are strongest.

Interpretation. The coefficient on exposure measures how the arrival of nearby garment work changes the annual hazard of marriage or childbearing. Because exposure is cumulative, it captures both direct work incentives and earlier schooling responses.

3.3 Educational attainment

For child i in family f in village v , the completed-schooling regression is:

$$\begin{aligned} \text{Education}_{ivf} = & \delta_1 \text{GarmentVillage}_v \times \text{YearsExposure}_{ivf} + \delta_2 \text{GarmentVillage}_v \times \text{YearsExposure}_{ivf} \times \text{Female}_{ivf} \\ & + \gamma_1 \text{GarmentVillage}_{ivf} \times \text{Female}_{ivf} + \gamma_2 \text{GarmentVillage}_{ivf} \times \text{YearOfBirth}_{ivf} \\ & + \gamma_3 \text{GarmentVillage}_{ivf} \times \text{YearOfBirth}_{ivf} \times \text{Female}_{ivf} + \lambda_1 \text{YearOfBirth}_{ivf} + \lambda_2 \text{YearOfBirth}_{ivf}^2 \\ & + \lambda_3 \text{Female}_{ivf} \times \text{YearOfBirth}_{ivf} + \lambda_4 \text{Female}_{ivf} \times \text{YearOfBirth}_{ivf}^2 + \theta_f + \varepsilon_{ivf}. \end{aligned}$$

- θ_f are sibling fixed effects in the main specification.
- δ_2 is the extra effect of garment exposure on girls relative to their brothers.
- $\delta_1 + \delta_2$ compares girls in garment villages to girls in non-garment villages.

Interpretation. This specification is designed to identify whether garment growth closes the gender education gap, not merely whether all education rises over time. Comparing sisters to brothers within the same family removes many household-level confounds.

3.4 School enrollment over the life cycle

To study when schooling changes occur, the authors estimate:

$$\begin{aligned} \text{Enroll}_{ivft} = & \delta_1 \text{GarmentVillage}_v \times \text{PostGarments}_{ivft} \\ & + \delta_2 \text{GarmentVillage}_v \times \text{PostGarments}_{ivft} \times \text{Female}_{ivft} \\ & + \delta_3 \text{GarmentVillage}_v \times \text{PostGarments}_{ivft} \times \text{Age}_{ivft} \\ & + \delta_4 \text{GarmentVillage}_v \times \text{PostGarments}_{ivft} \times \text{Female}_{ivft} \times \text{Age}_{ivft} \\ & + \gamma_1 \text{GarmentVillage}_{ivf} \times \text{Female}_{ivft} \\ & + \gamma_2 \text{GarmentVillage}_{ivf} \times \text{Year}_t + \gamma_3 \text{GarmentVillage}_{ivf} \times \text{Year}_t \times \text{Female}_{ivft} \\ & + \sum_{a=5}^{18} \beta_a 1(\text{Age} = a) + \sum_{a=5}^{18} \beta_{a,f} 1(\text{Age} = a) \text{Female}_{ivft} \\ & + \lambda_t + \lambda_t \times \text{Female}_{ivft} + \theta_f + \varepsilon_{ivft}. \end{aligned}$$

- Enroll_{ivft} equals one if the child is enrolled in school in year t .
- $\text{PostGarments}_{ivft}$ equals one once garment work has become available to the child's origin village by year t .

- δ_2 is the effect on young girls relative to brothers.
- δ_4 lets that effect vary with age.

Interpretation. This regression reveals the life-cycle pattern hidden in the attainment regression. The key question is whether factories pull older girls out of school or keep younger girls in school longer because returns to literacy and numeracy have risen.

3.5 Benchmarking against the female stipend program (FSP)

The paper compares garment expansion to Bangladesh’s female school stipend program.

For enrollment, the authors use a triple-difference design of the form:

$$\begin{aligned} \text{Enroll}_{ivft} = & \alpha_1 (\text{Post1994}_t \times 1[\text{Schooling} \geq 6]) \\ & + \alpha_2 (1[\text{Schooling} \geq 6] \times \text{Female}_{ivft}) \\ & + \alpha_3 (\text{Post1994}_t \times 1[\text{Schooling} \geq 6] \times \text{Female}_{ivft}) + \text{controls} + \varepsilon_{ivft}. \end{aligned}$$

- The program began in 1994.
- Eligibility starts at grade 6 and above.
- The triple interaction is the treatment effect of the stipend program on eligible girls.

For marriage and first birth, because boys are no longer a clean control group, the authors estimate a female-only double-difference using exposure defined by $\text{Post1994}_t \times 1[\text{Schooling} \geq 6]$.

Interpretation. The FSP comparison is not the main identification strategy of the paper; it is a **benchmark**. The point is to ask whether a famous girls’ schooling policy explains Bangladesh’s educational change better than manufacturing growth does.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the design

- **Across-village variation.** Girls in garment-proximate villages are compared to girls in nearby non-garment villages.
- **Within-garment-village timing variation.** The first year in which anyone from the village worked in garments differs across villages, so cohorts within the same type of village face different exposure lengths.
- **Within-family variation.** Sibling fixed effects compare girls to their brothers, helping net out household-level factors.

4.2 Why factory timing creates usable quasi-experimental variation

- Factory owners report choosing locations mainly because of proximity to roads/infrastructure and the availability of family-owned buildings.

- Given the 2014 supplemental survey, the authors can measure not just whether a village is near factories, but **when** garment exposure began.
- This timing variation is central because the main outcomes depend on **age-specific exposure**.

4.3 Main threats and the paper's responses

- **Non-random location of factories.** Garment villages are closer to Dhaka and somewhat different at baseline.
 - Response: include garment-village indicators and differential trends; rely on timing variation; use sibling fixed effects where possible.
- **Different pre-trends.**
 - Response: Table 5 shows no statistically significant differential pre-trends in enrollment, marriage, or first birth before factories arrive.
- **Omitted infrastructure changes.**
 - Response: the paper argues it is hard for a single omitted factor to explain the whole pattern: strong effects for girls but not boys, larger effects on younger girls, and schooling-specific responses.
- **Selective migration or selective residence.**
 - Response: the retrospective modules collect information on children even if they no longer live in the household.

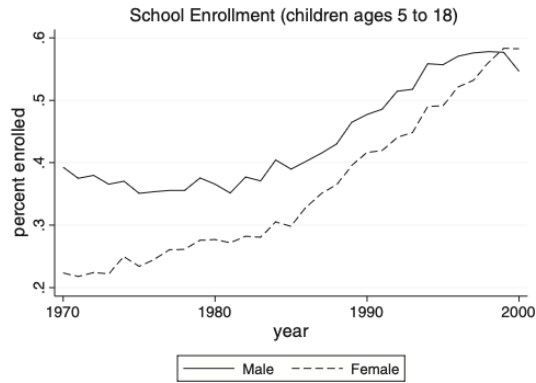
4.4 What the paper does *not* do

- There is **no instrument** for factory access.
- The paper is therefore a careful reduced-form difference-in-differences study, not a structural model of schooling or marriage choice.
- Some residual concerns remain, especially because factory placement is clearly related to geography. The paper's credibility rests on the cohort-timing design, sex comparisons, and pre-trend evidence.

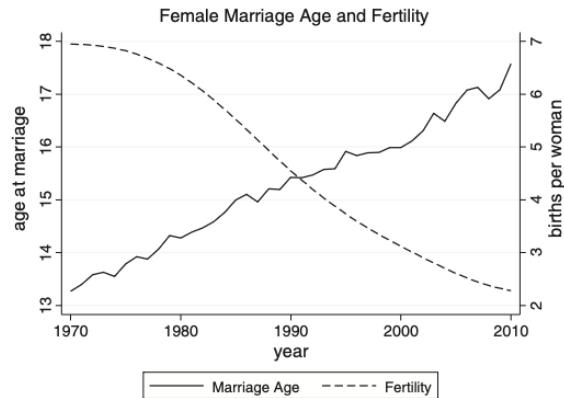
4.5 Bottom line on identification

The paper's empirical logic is persuasive because the results line up across multiple outcomes and mechanisms:

1. exposure raises girls' work,
2. exposure delays girls' marriage and fertility,
3. exposure raises girls' schooling more than boys', and
4. the effects are concentrated at ages where those channels should matter.



Source: Bangladesh Household Income and Expenditure Survey



Source: World Bank Development Indicators (fertility), Demographic and Health Surveys (marriage age)

Fig. 1. Enrollment, female marriage age, and fertility trends in Bangladesh. Note: Age at marriage is the average age of marriage of women entering their first marriage in the year given on the x-axis. Fertility represents the number of children that would be born to a woman if she were to live to the end of her childbearing years and bear children in accordance with current age-specific fertility rates, as calculated by the World Bank for the World Development Indicators: <http://data.worldbank.org/indicator/SP.DYN.TFRT.IN>.

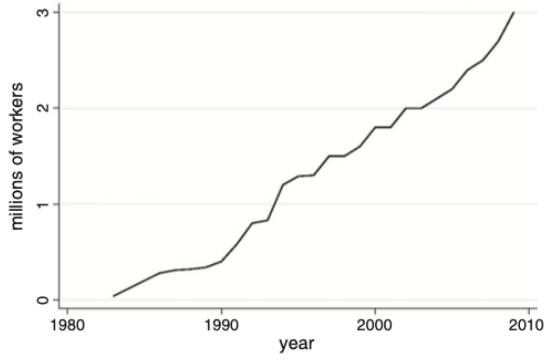
5 Tables and figures

5.1 Figure 1: national trends in enrollment, marriage age, and fertility

- Girls' enrollment rises sharply and catches up to or exceeds boys' enrollment by the end of the sample horizon.
- Female marriage age rises from about 13 to above 17.
- Fertility falls dramatically from about 7 births per woman to near 2.
- This figure motivates the paper: something large changed in girls' lives in Bangladesh, and the paper asks whether garment growth explains part of it.

5.2 Figure 2 and Figure 3: garment-sector growth and worker characteristics

- Figure 2 shows the explosive expansion of national garment employment.



Source: Bangladesh Garments Manufacturers and Exporter Association (BGMEA)

Fig. 2. Nation-wide employment in the garment industry.
Source: Bangladesh Garments Manufacturers and Exporter Association (BGMEA).

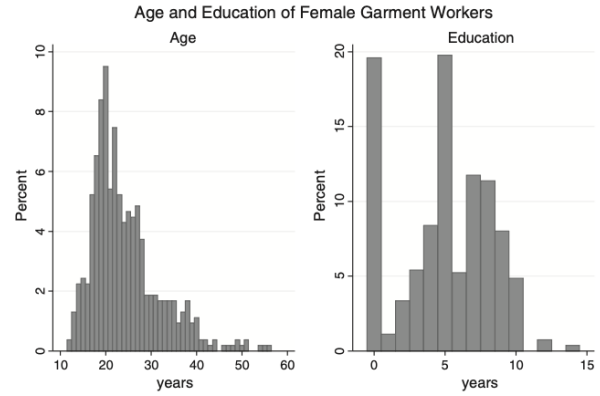


Fig. 3. Characteristics of sampled female garment workers.

- Figure 3 shows that sampled female garment workers are mostly young and are more educated than the average woman in the surrounding villages.
- The visual message is that garment jobs are a realistic alternative to early marriage for teenage and young adult women, and that these jobs reward basic schooling.

5.3 Table 1 and Table 2: sample descriptives and baseline differences

Table 1
Summary statistics of individuals in 2009.

	Garment Villages			Overall	Non-Garment Villages		
	Garment Workers	Workers Not in Garment Factories	Non-Workers		Workers	Non-Workers	Overall
Age	26.137 [0.266]	33.896 [0.449]	30.190 [0.196]	29.901 [0.253]	30.291 [0.799]	33.256 [0.395]	28.741 [0.360]
Female	0.541 [0.016]	0.174 [0.014]	0.539 [0.589]	0.402 [0.008]	0.144 [0.021]	0.527 [0.016]	0.600 [0.010]
Wage (Taka) ^a	3945 [1111]	5992 [1441]	4363 [1111]	4363 [1111]	3900 [87.0]	3900 [199]	3900 [199]
Years of experience ^c	4.226 [0.137]	12.056 [0.465]	7.909 [0.465]	7.909 [0.750]	12.189 [0.750]	12.189 [0.750]	12.189 [0.750]
Years of education	6.054 [0.114]	4.884 [0.160]	4.363 [0.342]	4.904 [0.064]	3.895 [0.224]	3.309 [0.110]	3.250 [0.095]
Mother's years of education	1.534 [0.086]	1.06 [0.088]	1.826 [1.826]	1.595 [0.046]	0.607 [0.097]	1.094 [0.073]	0.868 [0.044]
House has a cement floor	0.896 [0.013]	0.936 [0.018]	0.616 [0.016]	0.668 [0.008]	0.119 [0.019]	0.296 [0.013]	0.221 [0.011]
Married	0.749 [0.014]	0.818 [0.014]	0.596 [0.596]	0.681 [0.008]	0.554 [0.029]	0.541 [0.015]	0.531 [0.009]
N	965	753	1998	3716	258	1029	1287

^a Conditional on working. The standard error of each mean in brackets below. Sample includes all members of households age 7+ in 2009 survey.

Table 2
Differences in garment vs. non-garment villages.

	Garment villages	Non-garment villages	P-value for diff.	N _{garment}	N _{non-garment}
Individual level					
Completed education, females 50+	0.824	0.537	0.232	176	80
Completed education, males 50+	3.486	1.943	0.002***	222	88
Age at marriage, females 50+	14.788	14.462	0.604	85	39
Age at first birth, females 50+	19.286	21.162	0.080*	84	37
Village level					
Distance to Dhaka (km)	1.795	6.813	<0.001***	44	16
Distance to a girls' secondary school (km)	5.659	6.375	0.662	44	16
Distance to a boys' secondary school (km)	6.932	10	0.16	44	16
Male age: wage (peak season, in Taka)	27.559	27.997	0.802	44	16
Female age: wage (peak season, in Taka)	22.563	22.701	0.945	44	16

Only individuals born in village included.

* p < 0.1.

** p < 0.05.

*** p < 0.01.

- Table 1 shows garment workers are younger and more educated; among comparable women, garment work appears to pay better than other available work.
- Table 2 shows garment villages are closer to Dhaka and have somewhat better baseline schooling, especially for older men.
- These tables justify the paper's caution about raw comparisons and motivate the need for timing-based identification.

5.4 Figure 4, Figure 5, and Table 5: timing of exposure and pre-trends

- Figure 4 shows when villages first gained garment exposure; many villages enter in the 1980s and 1990s.
- Figure 5 shows substantial variation in years of exposure across the education and marriage samples.

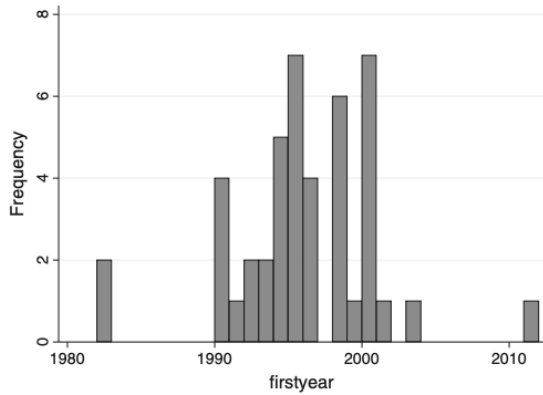


Fig. 4. Date in which people in each garment-proximate village first began working the garment industry.

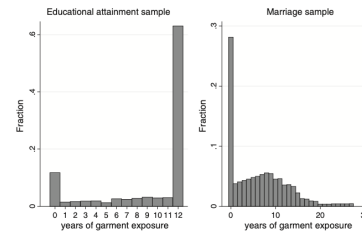


Fig. 5. Identifying variation in garment exposure. Note: The distribution of exposure in the childbearing regression is extremely similar to the sample in the marriage regression and thus omitted for the sake of brevity.

Table 5
Pre-garment industry trends in enrollment, marriage and time of first birth.

Sample	All	Females	Females
Dependent variable	Enrolled	Married	First birth
Garment village × Years to garment industry	0.00515		
	[0.00415]		
Garment village × Female × Years to garment industry	0.00295	6.84E−05	2.51E−11
	[0.00461]	[5.86E−05]	[4.95E−10]
Observations	6,214	21,610	22,406
Mean dependent variable	0.412	0.016	0.011

The sample in each regression is individuals in garment villages, in years before the first person from that village worked in a garment factory. Years to garment industry is calculated by subtracting the year in which people from that village first worked in a garment factory from the year of the observation. It equals zero for all individuals not in garment villages. Column 1 includes family fixed effects, female X year of birth dummies, and age trends by gender; standard errors in brackets are clustered at the level of the family. Columns 2 and 3 include year dummies and a quadratic in age; standard errors in brackets are clustered at the level of the individual. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

- Table 5 shows no differential pre-trends in enrollment, marriage, or childbirth before factories arrive.
- This is one of the most important credibility checks in the paper.

5.5 Table 3 and Table 4: work, marriage, and childbearing

- Table 3 shows that girls exposed during ages 10–29 or 10–23 are much more likely to have worked outside the home.
- Table 4 shows that garment exposure lowers the hazard of marriage and first birth for girls, but not for boys.
- The age interactions and Figure 6 show that marriage-delay effects are strongest at ages 12–18.
- These are the central results linking manufacturing access to women's life-course decisions.

5.6 Table 6, Table 7, Table 8, and Figure 7: schooling and mechanisms

- Table 6 shows strong gains in educational attainment, especially for girls relative to boys.

- Table 7 shows these schooling effects are not larger in families where mothers or older sisters work; if anything, the effects are larger where they do not. This argues against a pure income or household-time reallocation story.
- Table 8 and Figure 7 show that enrollment gains are concentrated among younger girls, while some dropout pressure may appear among older girls.
- The age profile is exactly what one would expect if factories raise returns to education for pre-working-age girls but raise the opportunity cost of school for girls near working age.

5.7 Table 9 and Table 10: the female stipend program comparison

- Table 9 finds no positive triple-difference effect of the stipend program on enrollment; the point estimate is negative.
- Table 10 finds no evidence that the stipend program delayed marriage or first birth.
- The authors' conclusion is that garment-sector expansion is a more plausible explanation for the closing of the gender schooling gap than the widely cited cash-transfer program.

Table 9

Effects of the female stipend program on school enrollment.
(Dependent variable = 1[Enrolled]).

Post 1994 × ≥6 years of school	−0.143*** [0.0439]	−0.153*** [0.0531]
≥6 years of school × Female	−0.0751 [0.0546]	−0.0215 [0.0691]
Post 1994 × ≥6 years of school × Female	−0.0097 [0.0609]	−0.0574 [0.0742]
Sibling FE's	Y	N
Observations	23,129	23,129
R-squared	0.058	0.058
Mean dependent variable	0.421	0.421
Mean FSP exposure	0.084	0.084

The triple interaction of post 1994 × ≥6 years of school × female defines exposure to the FSP. Dependent variable is enrollment in year t. Regressions include family fixed effects and controls for year fixed effects interacted with a female dummy, garment village × female, age fixed effects by gender, and the main effect of 6 years of schooling. Standard errors in brackets, clustered at the level of the family.

* p < 0.1.

** p < 0.05.

*** p < 0.01.

Table 10

Effects of the female stipend program on marriage and childbearing.

Dependent variable	Married	First birth
Post 1994 × ≥6 years of school	0.00107 [0.00132]	1.91E − 10 [3.20E − 09]
Observations	29,081	29,081
Mean dependent variable	0.0111	0.0111
Mean FSP exposure	0.1157	0.1259

Only females included. In the female-only sample, the interaction of post 1994 × ≥6 years of school defines exposure to the FSP. The dependent variable equals 1 in the year in which an individual was married or had his/her first child. Regressions include controls for age, age squared, dummies for year, and a dummy for garment village in columns 3 and 4. Standard errors in brackets, clustered at the level of the individual; ***p < 0.01, **p < 0.05, *p < 0.1.

6 Short summary

- **Method.** Reduced-form difference-in-differences and hazard models using retrospective household histories, village timing of factory access, and sibling fixed effects.
- **Identification.** Compare garment-proximate and control villages, exploit when garment work first became available, and use brothers as within-family controls for some schooling outcomes.
- **Application.** Rural/peri-urban Bangladesh during the rise of the garment industry.
- **Main result.** Garment-job access increased girls' labor force participation, delayed marriage and childbearing, and raised girls' schooling substantially; these effects appear stronger than those of the female stipend program in this setting.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The rapid expansion of low-skill manufacturing in Bangladesh significantly improved young women's outcomes.
- The garment industry delayed marriage and childbirth and increased girls' educational attainment.
- These effects work through both direct employment opportunities and increased expected returns to schooling.

7.2 Economic intuition

- For older girls, garment jobs raise the opportunity cost of marrying early because factory work becomes a realistic alternative.
- For younger girls, garment jobs raise the returns to literacy and numeracy, so parents keep daughters in school longer.
- The same industry can therefore produce opposite schooling forces at different ages: more enrollment for younger girls and some dropout pressure for older girls near labor-market entry.

7.3 Takeaway

This paper is a clean example of how **labor demand shocks can transform household decisions**. It shows that industrialization does not only raise incomes; it can also reshape the timing of marriage, fertility, and education for women. For Bangladesh, the paper's main lesson is that the rise of garment manufacturing was not just an export story — it was a major social change in the lives of girls and young women.